

System Design for High-Capacity Unrepeated Optical Transmission

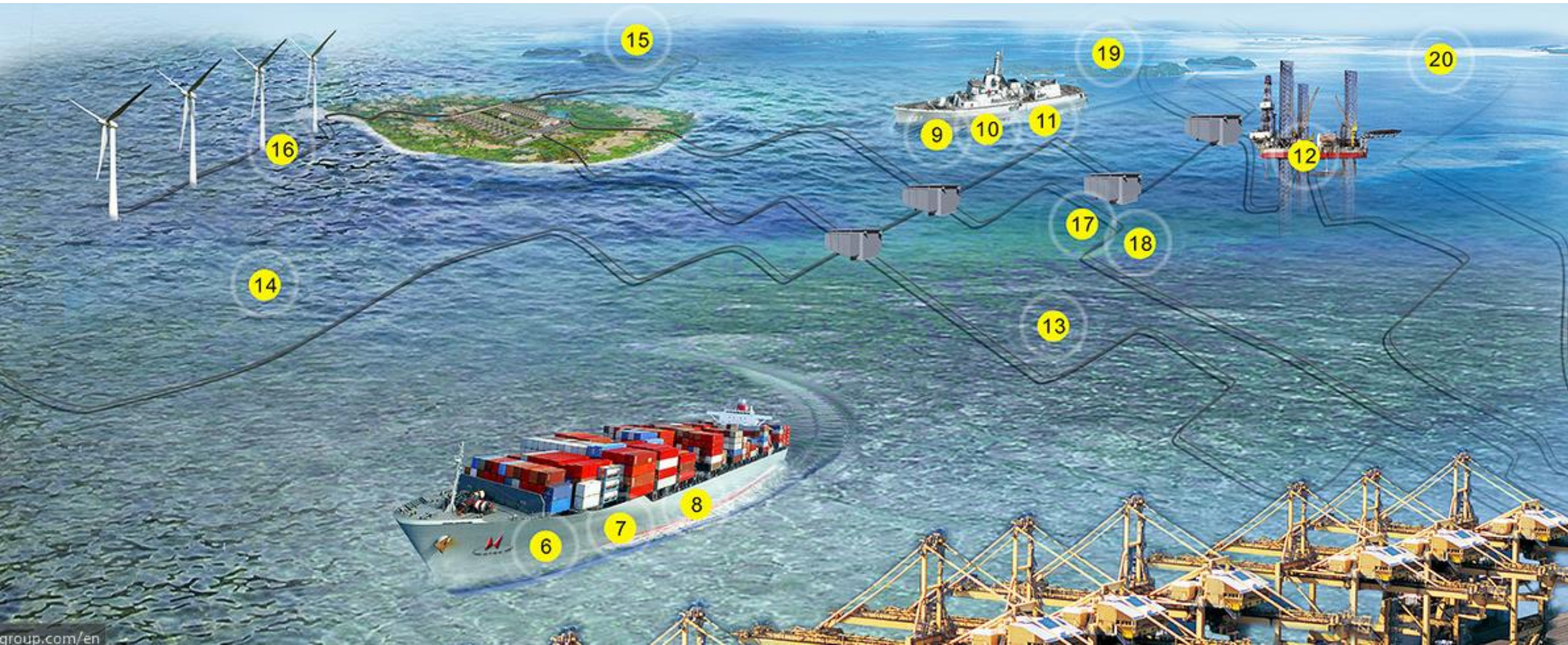
Ph.D. Thesis, João Carlos S. S. Januario

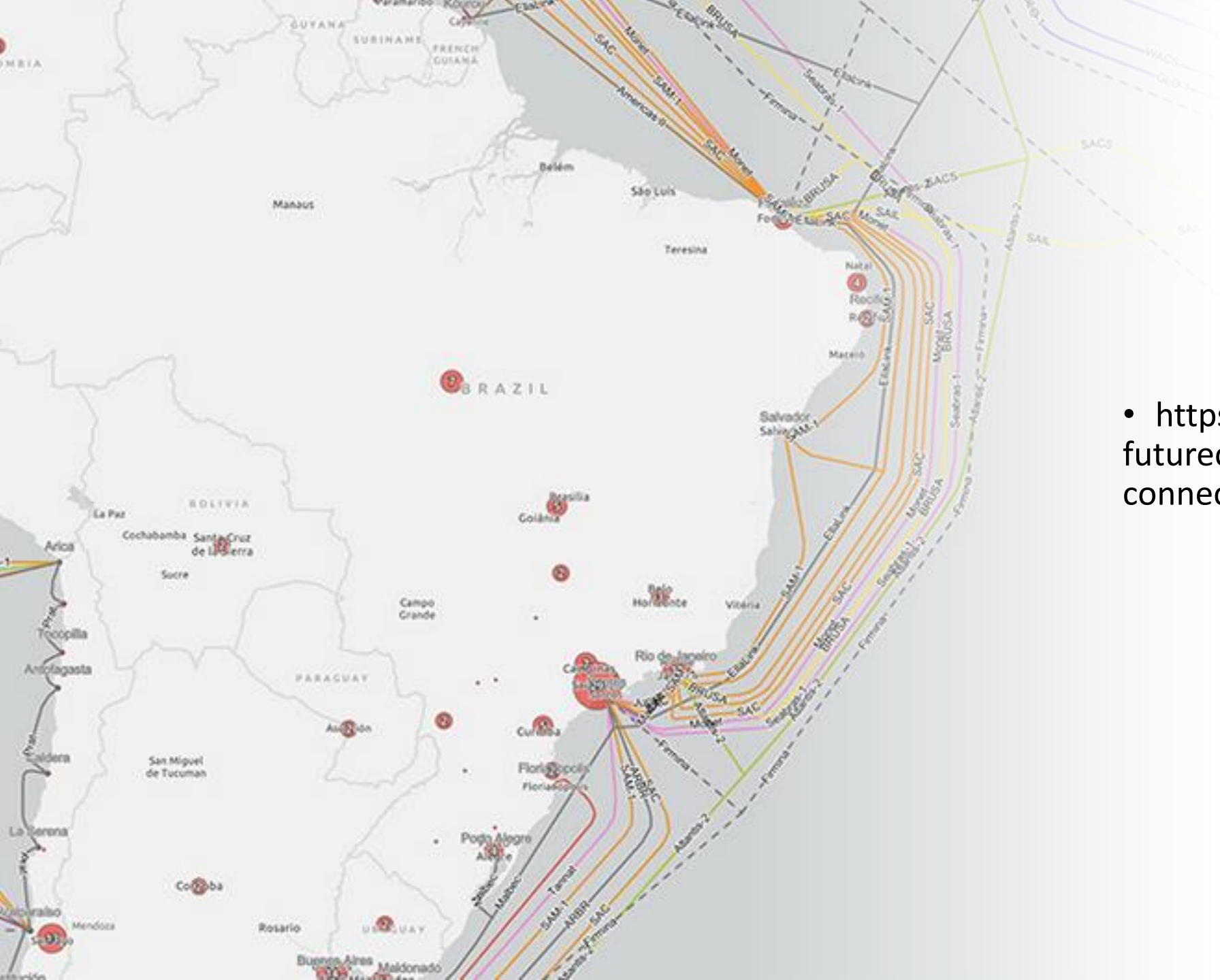
Unicamp, March 2023

Project “Amazonia Conectada”



Repeaterless Submarine Optical Cable





- <https://www.infinera.com/blog/futurecom-2022-a-celebration-of-connectivity/tag/metro/>

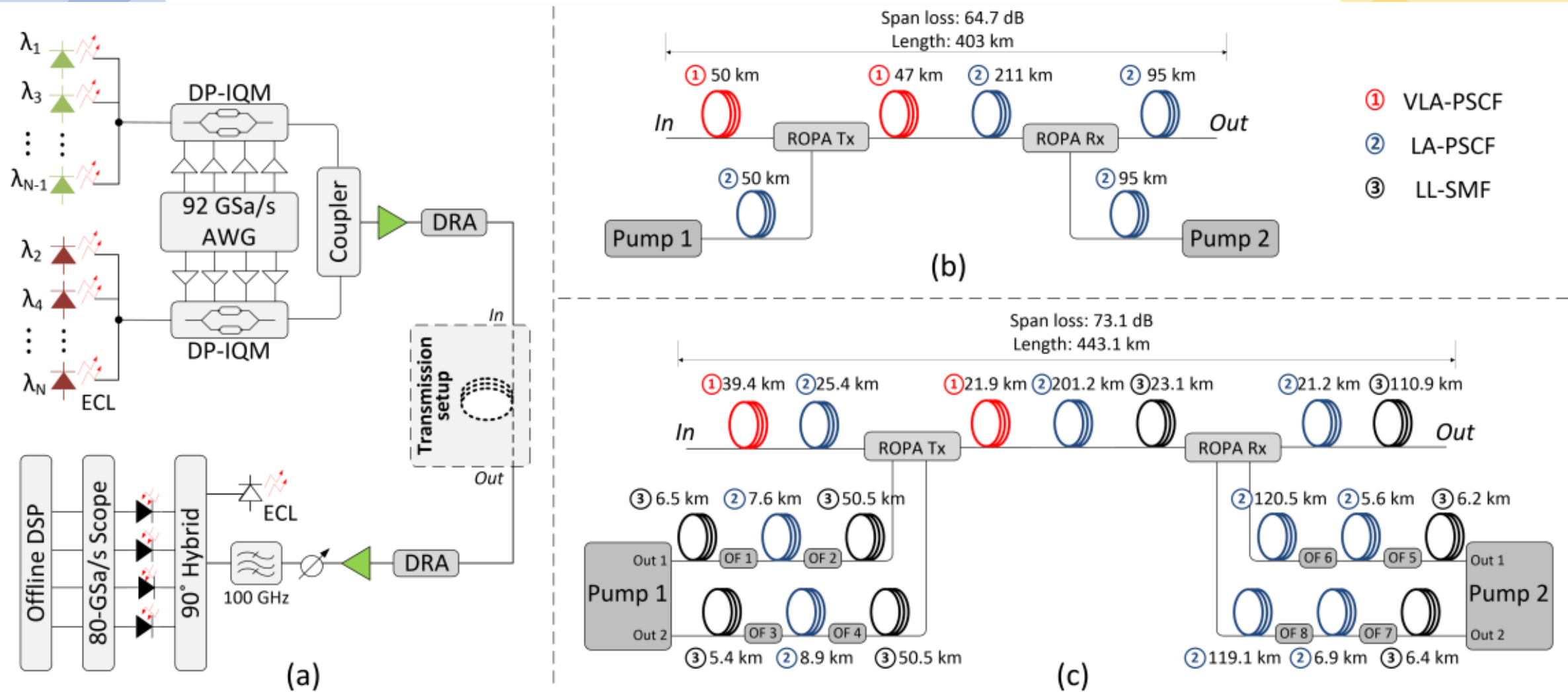


Fig. 1. Experimental arrangement: (a) transmitter and receiver stages; (b) 403-km hybrid transmission link with single-stage dedicated ROPA pumping; (c) 443.1-km hybrid link with dual-stage pumping. The transmission setups (b) and (c) depict the dashed box in (a).

Strategy to Improve Performance

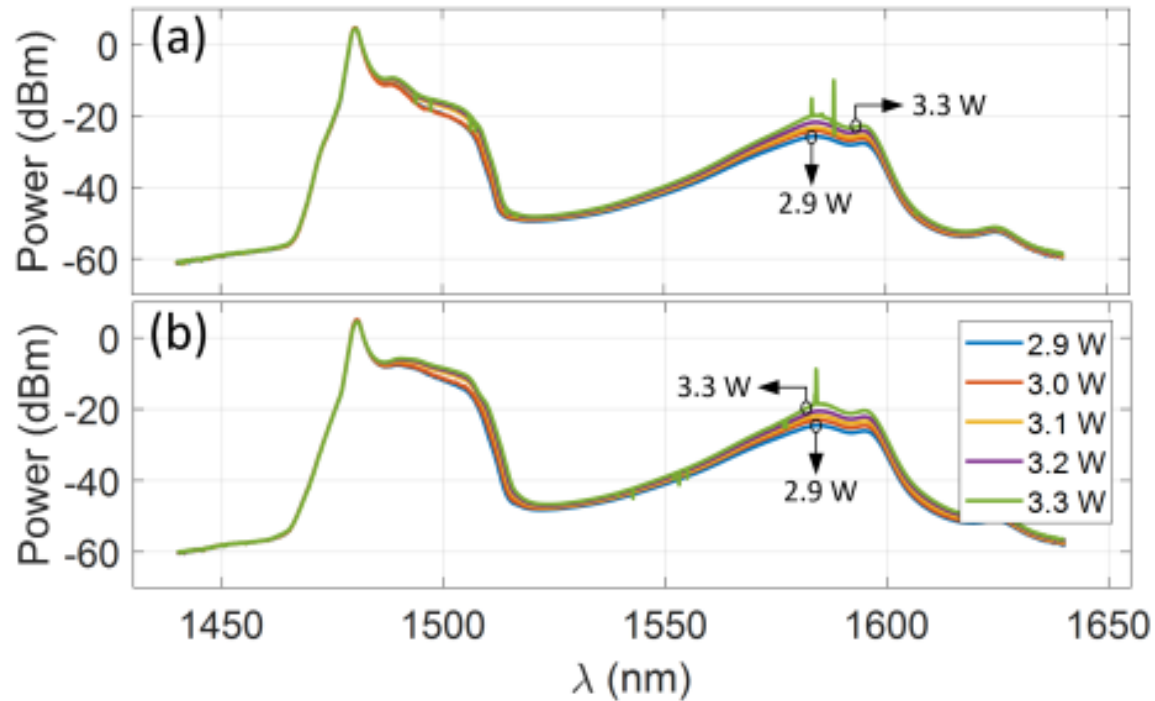


Fig. 4. Optical spectra at the output of ROPA Tx delivery link for the unit pump source out (a) 1 and out (b) 2. Lasing induced fluctuations can be observed at 3.3-W pump power.

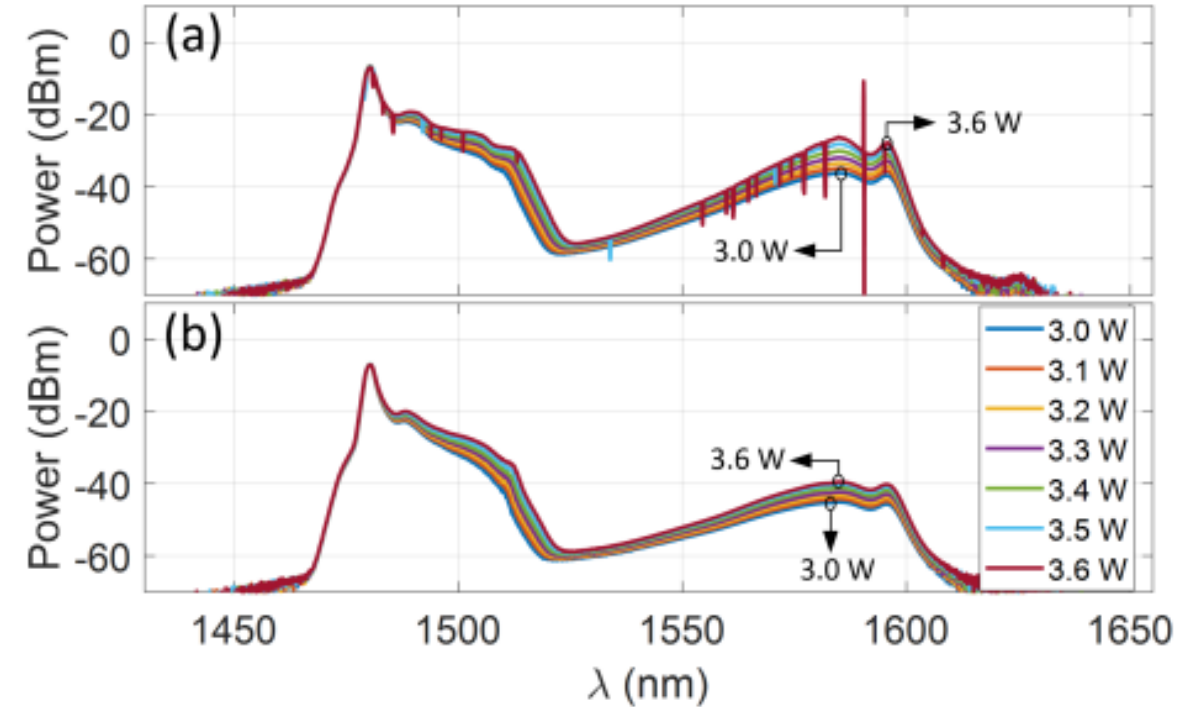


Fig. 5. Optical spectra at the output of ROPA Rx delivery link for the unit pump source out (a) 1 and out (b) 2. Instabilities begin to appear when the pump power level approaches 3.6 W.

Main Goal: 400 Gb/s @ 400 km

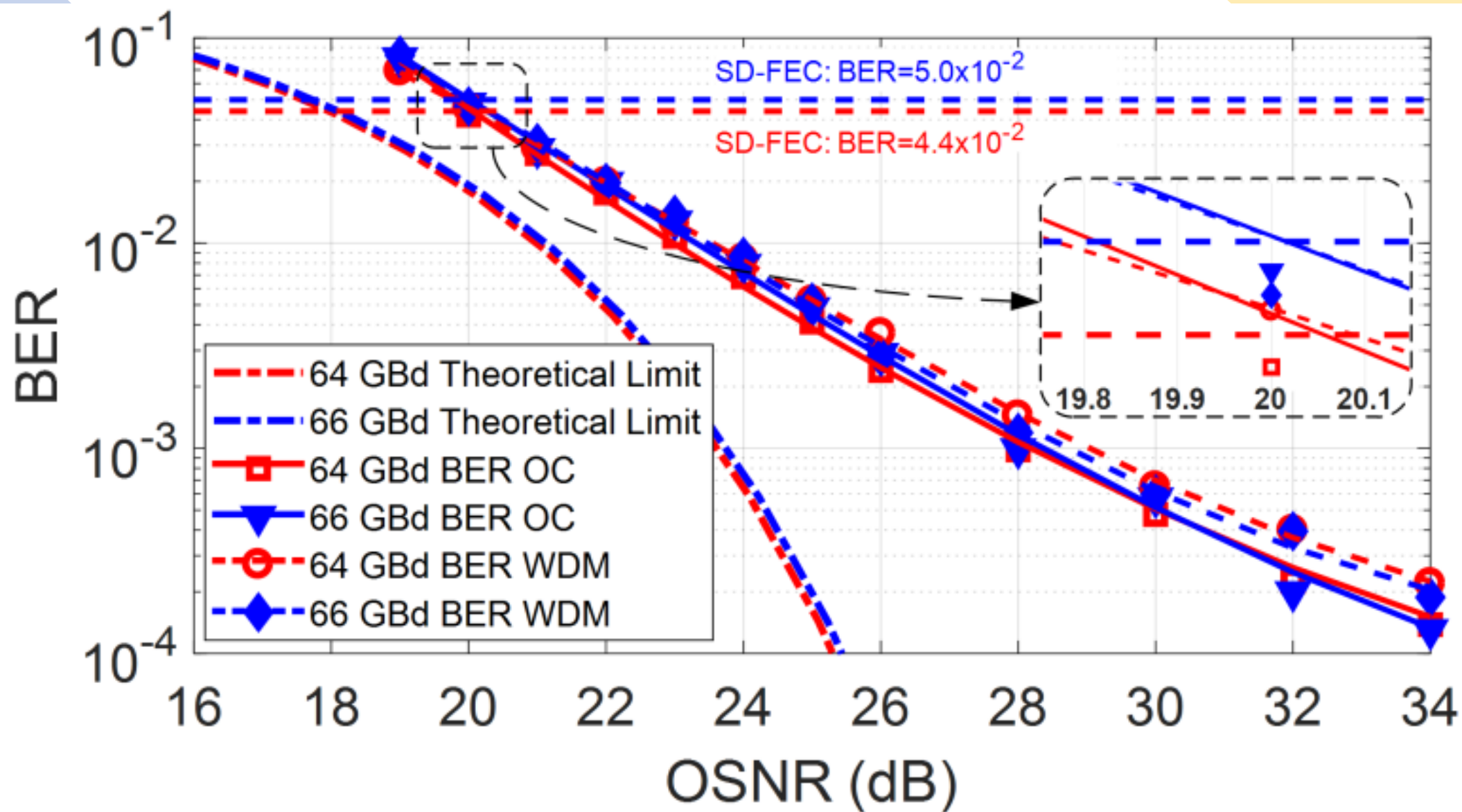


Fig. 8. BER vs. OSNR curves in B2B for 64 (red curves) and 66 GBaud (blue curves) symbol rates, considering one-channel (OC) and WDM configurations.